Confidence Interval

Formula:

$$CI = \bar{x} \pm z\left(\frac{\sigma}{\sqrt{n}}\right)$$

Text:

The confidence interval is used to estimate the range within which the true population mean is expected to lie. A 95% confidence interval indicates that the true mean BMI is likely to fall within the calculated range.

---

t-Test

Formula:

$$t = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$$

Text:

The t-test compares the mean BMI values between smokers and non-smokers to determine whether there is a statistically significant difference between the two groups.

Chi-Square Test

Formula:

$$\chi^2 = \sum \frac{(O - E)^2}{E}$$

Text:

The chi-square test measures the association between categorical variables such as smoking status and diabetes.

ANOVA Test

Formula:

$$F = \frac{\text{Variance Between Groups}}{\text{Variance Within Groups}}$$

Text:

$$H_1: \mu_1 \neq \mu_2$$

ANOVA is used to compare the BMI means of multiple age groups and determine whether at least one group differs significantly from the others.

Covariance

Formula:

$$Cov(X,Y) = \frac{\sum(X - \bar{X})(Y - \bar{Y})}{n - 1}$$

Text:

Covariance measures how two variables change together. A positive covariance between age and BMI indicates that both variables tend to increase together.

Correlation

Formula:

$$r = \frac{\sum(x - \bar{x})(y - \bar{y})}{\sqrt{\sum(x - \bar{x})^2 \sum(y - \bar{y})^2}}$$

Text:

Correlation measures the strength and direction of the relationship between age and BMI. Values closer to +1 indicate a strong positive relationship.

Null Hypothesis

Formula:

$$H_0: \mu_1 = \mu_2$$

Text:

The null hypothesis assumes that there is no significant relationship or difference between the selected variables.

Alternative Hypothesis

Formula:

$$H_1: \mu_1 \neq \mu_2$$

The alternative hypothesis assumes that a significant relationship or difference exists between the variables being analysed.